

71. Which of the following methods of data collection is mostly used in quantitative research?
- Observation
 - Questionnaire
 - Unstructured interviews
 - Key informant interviews
72. The last stage in research process is _____.
- Proposal development
 - Data collection and analysis
 - Report writing and presentation
 - Research topic selection
73. One of the following is NOT TRUE about the limitation of statistics
- Statistics is suitable to the study of quantitative phenomenon
 - Statistics table may be misused
 - Statistics does not study individuals
 - Statistics is only method of studying a problem
74. Which type of measure of central tendency is often used to get the most frequent value of the data?
- Mean
 - Mode
 - Median
 - Standard deviation
75. The arithmetic mean is 12 and the number of observations is 20, then the sum of all the values is _____.
- 8
 - 32
 - 240
 - 1.667
76. A box contains 10 balls out of which 2 are green, 5 are red and 3 are black. If two balls are drawn at random, one after the other without replacement, what is the probability that the first ball is red and the second one is black.
- $\frac{1}{6}$
 - $\frac{1}{9}$
 - $\frac{2}{9}$
 - $\frac{1}{15}$
77. The standard deviation of a sample of 100 observations equals to 64. The variance of the sample equals _____.
- 8
 - 1.56
 - 4.096
 - None of the above
78. If the null hypothesis (H_0) is true and we reject it is called _____
- Type-I Error
 - Type-II Error
 - Sampling Error
 - Standard Error

79. $1-\alpha$ is referred as _____.
 A. Power of test
 B. Confidence coefficient
 C. Level of significance
 D. Size of the test
80. A statistical analysis process by which we estimate the impact of one or more independent variables on the dependent variable is _____.
 A. Correlation
 B. Regression
 C. Residual
 D. Slope
81. An operations manager makes decision on "the unique features of company products that satisfy customer needs." This type of decision represents _____.
 A. Tactical decision
 B. Strategic decision
 C. Productivity decision
 D. Operational decision
82. Goods and service design focus can slightly be different. One of the following is INCORRECT regarding service design.
 A. Service design - Designs value conveyed as used
 B. Service design - Designs can be often standardized
 C. In Service design - Quality is inherent in the process
 D. Service design - Set standard for performance
83. Which of the following is FALSE about job shop processes?
 A. Makes small numbers of a wide variety of products.
 B. Job shops have high capital costs and high unit costs
 C. One example of job shops are restaurants
 D. Uses a general purpose equipment
84. A big advantage of a process-oriented layout is;
 A. Its flexibility in equipment and labour
 B. Its low cost
 C. The ability to employ low-skilled labour
 D. Its high equipment utilization
85. Which of the following is ALWAYS TRUE when production systems move from projects to job shop to batch production to mass production to continuous production process?
 A. Processes become more flexible
 B. Customer involvement with the process increases
 C. Products become more standardized
 D. Demand volumes decrease

86. Which of the following is FALSE about capacity?
- Capacity is usually purchased in chunks
 - Capacity is not tactical issue but strategic issue
 - Capacity is the maximum output rate of a facility
 - Capacity planning is necessary to meet demand
87. Which of the following is NOT the ultimate goal of scheduling in operations process?
- Efficient utilization of inventories
 - Efficient utilization of staff
 - Efficient utilization of facility
 - Minimization of customer waiting time
88. All of the following costs are likely to decrease as a result of better quality management, EXCEPT;
- Customer satisfaction
 - Inspection costs
 - Maintenance costs
 - Warranty and service costs
89. As an area of study, financial management is concerned with _____.
- the financing function
 - the forecasting function
 - the demand creation function
 - distribution of profit and losses function
90. Net profit of a company is Br. 360,000 and total sales of the company is Br. 3,000,000, Net profit ratio of the company will be _____.
- 160%
 - 12%
 - 10%
 - 100%
91. What is the total amount accumulated after three years if someone invests Br. 1,000 today with a compound annual interest rate of 5 percent?
- Br. 1,150
 - Br. 1,110
 - Br. 1,103
 - Br. 1,158
92. Mr. Anteneh purchased 100 stocks of Ashewa IT solutions ltd, for Br.21 on March 15, 2020 and sold for Br.35 on March 14 next year. In the company paid a dividend of Br.2.50 per share, then Anteneh holding period return is _____.
- 11.90%
 - 45.40%
 - 66.70%
 - 78.60%

93. A project costs Br. 1,00,000 and annual cash flow of Br. 20,000 for 8 years. It's payback period is ____.
- A. 1 year
B. 2 years
C. 3 years
D. 5 years
94. Which of the following is NOT a common area of application of operations research techniques in business decisions?
- A. Marketing and advertising
B. Production decisions
C. Human resource assignment
D. Job analysis and job description
95. In linear programming problems, ____ are restrictions or limits coming from a variety of sources.
- A. Decision variables
B. Objective function
C. Constraints
D. Parameters
96. Which of the following CANNOT be an objective of transportation problem?
- A. To identify the optimal shipping routes
B. To identify the maximum amount that can be shipped over the optimum route
C. To determine the total transformation cost or the profit of transportation
D. To determine the total profit from the product
97. Which of the following assumption is NOT TRUE about assignment problems?
- A. Number of jobs is equal to the number of machines or persons
B. Each man or machine is loaded with one and only one job
C. Each man or machine is independently capable of handling any of the job given
D. All employees are assumed to have the same performance level
98. In a payoff table, if the row maximum values are given 15, 9 and 3, what will be the decision in using MAXIMAX decision criterion?
- A. 15
B. 9
C. 3
D. 13.5
99. Which statement is NOT TRUE about MAXIMIN decision criterion?
- A. This principle is adopted by *pessimistic* decision makers who are conservative in their approach.
B. The decision maker assumes that the worst outcome will occur.
C. It is one approach under complete uncertainty decision making condition
D. It is often used by an optimist decision maker